



مدرس المادة: شادي الشوافقة
موقع المحاضرة: ١١.١٠ هذه المرة خم

Answer the following five questions:

1- Let X and Y be two vectors in $\mathbb{R}^n$. Prove that

Hint: $|X \cdot Y| \leq \|X\| \|Y\|$.

2- Verify the triangle inequality for $X = [1 \ 3 \ 2 \ -1]^T$, $Y = [1 \ 0 \ -3 \ 2]^T$

3- Let $X = x_1\hat{i} + x_2\hat{j} + x_3\hat{k}$, $Y = y_1\hat{i} + y_2\hat{j} + y_3\hat{k}$, $Z = z_1\hat{i} + z_2\hat{j} + z_3\hat{k}$
Show that

4- Find the area of the triangle with vertices X_1, X_2, X_3 where
 $X_1 = (1, -2, 4)$, $X_2 = (-3, 1, 4)$, $X_3 = (0, 4, 3)$

5- Let V be the set of all continuous function on $(0, \infty)$ and W be the set of all continuous functions on $[1, b]$. Prove that W is subspace of the vector space V under $f \oplus g = (f + g)(x)$, and $\alpha \odot f = (\alpha f)(x) \forall f, g \in V$, scalar α

Good Luck

SHADI SHAWAQFEN

Prob. C.5: Answer by True (T) or False (F):

1. If A is an $n \times n$ nonsingular matrix, then the row-echelon form of A is I . ()
2. If A is a singular $n \times n$ matrix, and B results by interchanging two rows of A , then B is singular. ()
3. If $B = 2A$, then $B^{-1} = 2A^{-1}$. ()
4. If A is a 4×4 orthogonal matrix, then $\text{rank}(A) < 4$. ()
5. If A and B are $n \times n$ matrices, then $(A + B)^T = A^T + 2AB + B^T$. ()
6. If A and B are $n \times n$ matrices, then $(A + B)(A + B) = A^2 + B^2$. ()
7. If A is a 4×4 matrix such that the homogeneous system $AX = 0$ has only the trivial solution, then $\text{rank}(A) < 4$. ()
8. If $\det(A) = 2$, then $\det(A^{-1}) = 0.5$. ()
9. If A is an $n \times n$ nonsingular matrix, then the columns of A span $\mathbb{R}^n$. ()
10. If the row-echelon form of A has a row of zeros, then there exist nontrivial solutions. ()
11. If A is an $n \times n$ singular matrix, then the system $AX = B$ has exactly one solution for every B . ()
12. If $A^{-1} = A^T$, then A is orthogonal. ()

Prob. C.6: What conditions must b_1 , b_2 , and b_3 satisfy in order the following system of equations to be consistent?

$$x + y + 2z = b_1$$

$$2x + y + 3z = b_2$$

$$x + z = b_3$$

Prob. C.7: Consider the set V of all vectors in $\mathbb{R}^3$ and define the operations by

$$X \oplus Y = X + 2Y$$

$$cX = 3cX$$

Determine whether the following operations are satisfied.

$$1. X \oplus (Y \oplus Z) = (X \oplus Y) \oplus Z$$

$$2. X \oplus (-X) = 0$$

$$3. c(X \oplus Y) = (cX) \oplus (cY)$$

$$4. (c + d)X = (cX) \oplus (dX)$$

1.

2.

3.

4.

Prob. C.8: Consider the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(t) & \sin(t) \\ 0 & -\sin(t) & \cos(t) \end{bmatrix}$

Using factor expansion find the inverse of the matrix A

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امتحان نایل

* سوال بیضی و مربع و نمودار [area of parallelogram]
Volume

* سوال بیضی و مربع و نمودار

1. The Polynomial
2. The eigenvalues of A is
3. The Rank of matrix
4. The inverse of matrix

* سوال [صیغ و خط] 20 علامه

* سوال حل (ای موضوع و نمودار نایل اسیر)

Let $x_1 = (1, 1, 1)$ $x_2 = (0, 1, 1)$
 $x_3 = (1, 0, 1)$
 $S = \{x_1, x_2, x_3\}$

PKE-Fet

1) Is S a basis of $\mathbb{R}^3$? why?

2)

أسئلة سنوات سابقة

المادة: جبر خطي

Q1- Find Basis For Nullspace of $A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}$

Q2- Find Basis For column space of $A = \begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ -1 & -6 & 2 \end{bmatrix}$

Q3- if $AB=C$ where $B = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & 2 \end{bmatrix}$,

$C = \begin{bmatrix} -5 & -1 & 0 \\ 6 & -3 & 7 \end{bmatrix}$ Find A.

Q4- $V = \{ (x, y, z) / x, y, z \in \mathbb{R} \}$, Define

$$(x_1, y_1, z_1) \oplus (x_2, y_2, z_2) = (x_1 + x_2, y_1 - y_2, z_1 + z_2)$$

give counter example to show that V is not vector space.

Q5- $T: P_1 \rightarrow \mathbb{R}^2$ is linear transform

$$T(2t+1) = (-2, 1) \text{ and } T(t-2) = (3, 2)$$

Find $T(5t-10)$.